INCH-POUND

MIL-DTL-83723/89F
1 October 2010
SUPERSEDING
MIL-DTL-83723/89E
23 May 2003

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, (CIRCULAR, ENVIRONMENT RESISTING), RECEPTACLE, (SINGLE HOLE MOUNT, THREADED COUPLING, SOLDER PIN CONTACT), (SERIES III, CLASSES H, J, L, N, P AND Y)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83723.

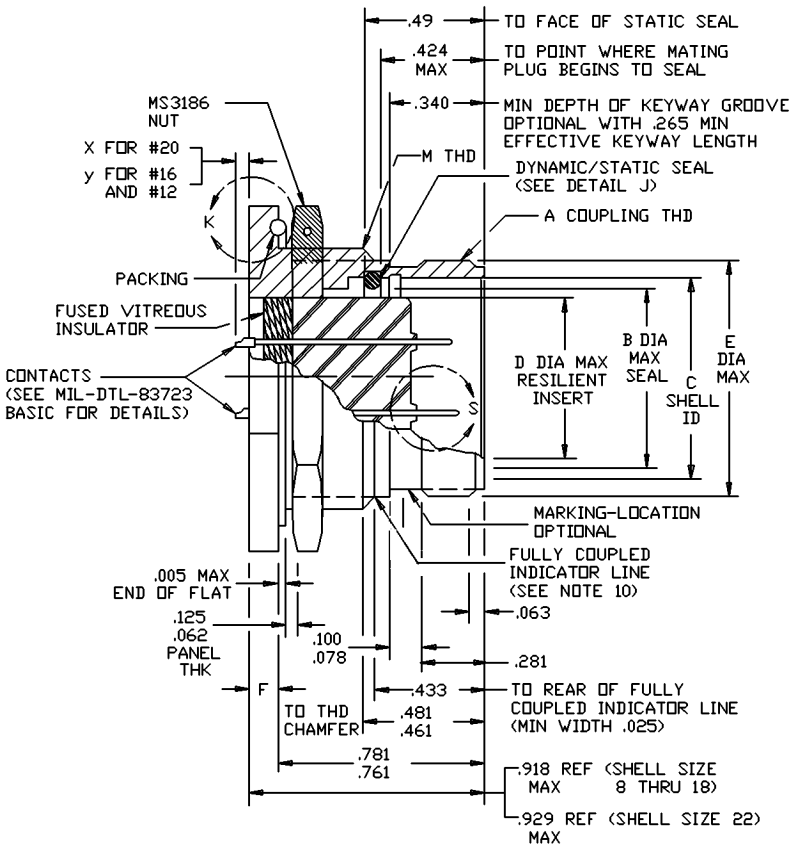


FIGURE 1. Dimensions and configurations.

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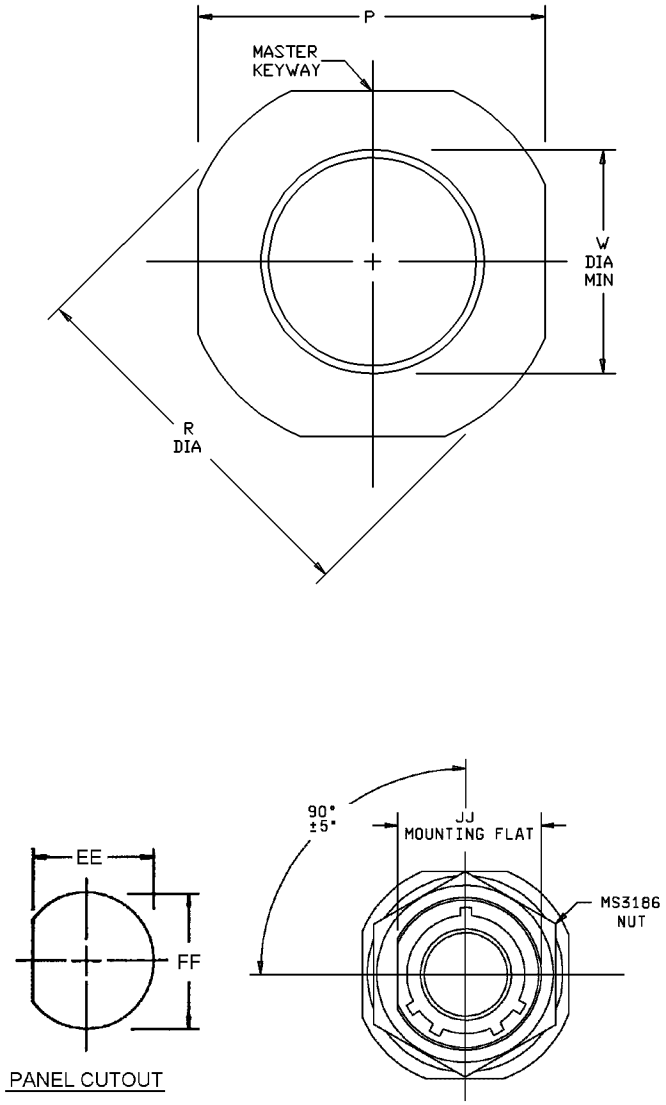
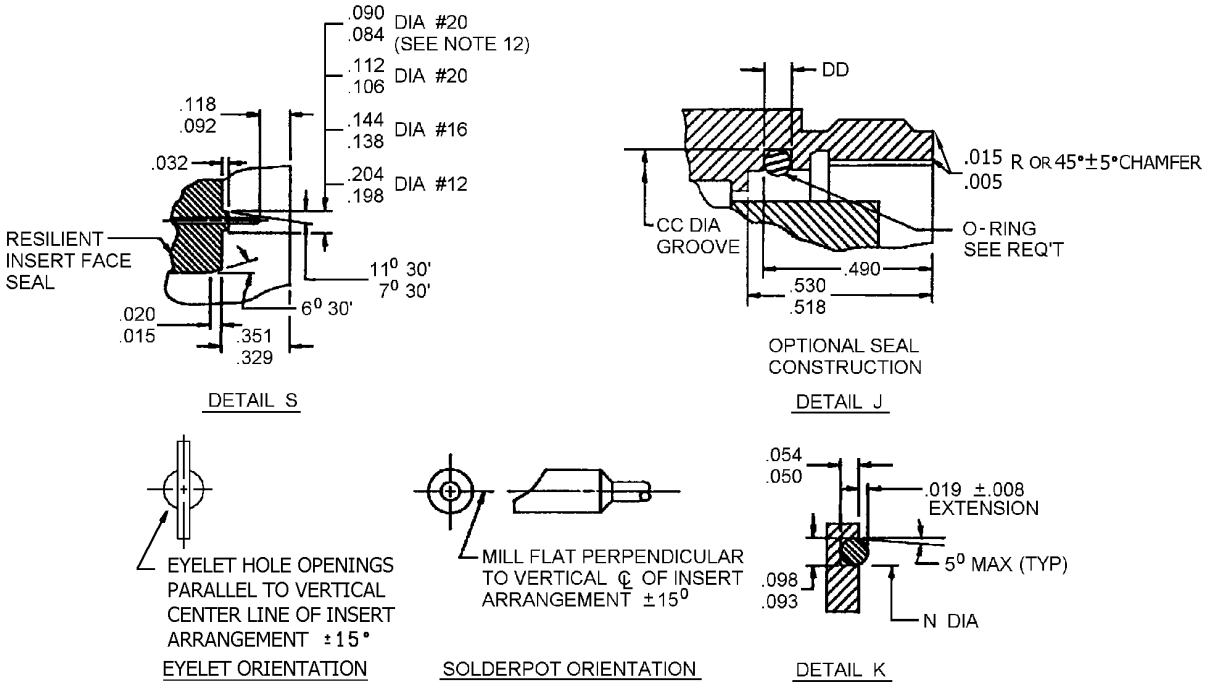


FIGURE 1. Dimensions and configurations – Continued.

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Shell size	A thread	B	C	D	E	F
8	.562-24 UNEF-2A	.394	.418 / .413	.290	.562	.137 / .097
10	.688-24 UNEF-2A	.515	.535 / .530	.388	.696	.137 / .097
12	.875-20 UNEF-2A	.685	.700 / .705	.558	.875	.137 / .097
14	.938-20 UNEF-2A	.754	.774 / .769	.627	.936	.137 / .097
16	1.062-18 UNEF-2A	.881	.901 / .896	.772	1.062	.137 / .097
18	1.188-18 UNEF-2A	.987	1.007 / 1.002	.860	1.187	.137 / .097
22	1.438-18 UNEF-2A	1.237	1.257 / 1.252	1.110	1.437	.148 / .128

Shell size	M thread	N	P	R	W	X
8	.625-20 UN-2A	.670	.980 / .949	1.078 / 1.047	.400	.180 / .120
10	.750-20 UNEF-2A	.796	1.104 / 1.073	1.192 / 1.162	.411	.180 / .120
12	.938-20 UNEF-2A	.984	1.291 / 1.260	1.380 / 1.349	.581	.180 / .120
14	1.000-20 UNEF-2A	1.046	1.391 / 1.360	1.516 / 1.485	.650	.180 / .120
16	1.125-18 UNEF-2A	1.171	1.516 / 1.485	1.641 / 1.610	.777	.180 / .120
18	1.250-18 UNEF-2A	1.296	1.641 / 1.610	1.766 / 1.735	.864	.180 / .120
22	1.500-18 UNEF-2A	1.609	1.954 / 1.923	2.078 / 2.047	1.133	.169 / 1.09

FIGURE 1. Dimensions and configurations – Continued.

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Shell size	Y	CC + .005 - .000	DD	EE	FF	JJ
8	.210 / .150	.490	.076 / .071	.605	.635	.596 / .590
10	.210 / .150	.636	.089 / .084	.730	.760	.721 / .715
12	.210 / .150	.806	.089 / .084	.917	.947	.908 / .902
14	.210 / .150	.875	.089 / .084	.980	1.010	.971 / .965
16	.210 / .150	1.002	.089 / .084	1.105	1.135	1.096 / 1.090
18	.210 / .150	1.108	.089 / .084	1.230	1.260	1.220 / 1.214
22	.199 / .139	1.358	.089 / .084	1.480	1.510	1.470 / 1.464

NOTES: (Notes 4 through 9 applicable to contacts only).

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.005 inches (0.13 mm) for three place decimal, ± .01 inches (0.25 mm) for two place decimals, and ± 1° for all angles.
4. Contact finish: Gold plate .000050 inch **(**0.00127 mm) minimum in accordance with ASTM B488. A suitable underplating shall be used. Silver shall not be used. Finish is applied after sealing.
5. Dimensions apply after plating.
6. Surface finish: 63 **(** $\sqrt{63}$) all over in accordance with ASME-B46.1 , unless otherwise specified.
7. Remove all burrs and sharp edges .003 to .005 inches (0.076 to 0.127 mm) R.
8. Diameters to be concentric within .004 inches (0.102 mm) TIR.
9. Diameters in the same cross-sectional plane to be concentric with each other .003 inches (0.076 mm) TIR.
10. Coupling indicator line shall be color coded red.
11. Shell size 8 will not mate with MIL-DTL-26500 connectors.
12. See table I for contact cavities requiring this dimension.

FIGURE 1. Dimensions and configurations – Continued.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.000050	0.00127	.137	3.48	.558	14.17	.901	22.89	1.192	30.28
.003	0.08	.138	3.51	.562	14.27	.902	22.91	1.214	30.84
.004	0.10	.139	3.53	.581	14.76	.908	23.06	1.220	30.99
.005	0.13	.144	3.66	.590	14.99	.917	23.29	1.230	31.24
.015	0.38	.148	3.76	.596	15.14	.918	23.31	1.237	31.42
.018	0.46	.150	3.81	.605	15.37	.929	23.60	1.250	31.75
.019	0.48	.169	4.29	.627	15.93	.936	23.77	1.252	31.80
.020	0.51	.180	4.57	.635	16.13	.947	24.05	1.257	31.93
.022	0.56	.198	5.03	.636	16.15	.949	24.10	1.260	32.00
.025	0.64	.199	5.05	.650	16.51	.965	24.51	1.291	32.79
.032	0.81	.204	5.18	.670	17.02	.971	24.66	1.296	32.92
.050	1.27	.210	5.33	.685	17.40	.980	24.89	1.349	34.26
.054	1.37	.265	.673	.696	17.68	.984	24.99	1.358	34.49
.062	1.57	.281	7.14	.700	17.78	.987	25.07	1.360	34.54
.063	1.60	.290	7.37	.705	17.91	1.002	25.45	1.380	35.05
.071	1.80	.329	8.36	.708	17.98	1.007	25.58	1.391	35.33
.076	1.93	.340	8.64	.715	18.16	1.010	25.65	1.437	36.50
.078	1.98	.351	8.92	.721	18.31	1.046	26.57	1.464	37.19
.084	2.13	.388	9.86	.730	18.54	1.047	26.59	1.470	37.34
.089	2.26	.394	10.01	.760	19.30	1.062	26.97	1.480	37.59
.090	2.29	.400	10.16	.761	19.33	1.073	27.25	1.485	37.72
.092	2.34	.411	10.44	.769	19.43	1.078	27.38	1.510	38.35
.093	2.36	.413	10.49	.772	19.61	1.090	27.69	1.516	38.51
.097	2.46	.418	10.62	.774	19.66	1.096	27.84	1.572	39.93
.098	2.49	.424	10.77	.777	19.74	1.104	28.04	1.609	40.87
.100	2.54	.433	11.00	.781	19.84	1.105	28.07	1.610	40.89
.106	2.69	.461	11.71	.796	20.22	1.108	28.14	1.641	41.68
.109	2.77	.481	12.22	.806	20.47	1.110	28.19	1.735	44.07
.112	2.84	.490	12.45	.860	21.84	1.133	28.78	1.766	44.86
.118	3.00	.515	13.08	.864	21.95	1.135	28.83	1.923	48.84
.120	3.05	.518	13.16	.875	22.23	1.162	29.51	1.954	49.63
.125	3.18	.530	13.46	.881	22.38	1.171	29.74	2.047	51.99
.128	3.25	.535	13.59	.896	22.76	1.187	30.15	2.078	52.78

FIGURE 1. Dimensions and configurations – Continued.

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REQUIREMENTS

Dimensions and configurations: See figure 1.

This connector mates with MIL-DTL-83723/86, /91, /95, and /97.

Insert arrangements and polarizing positions: See MIL-STD-1554 and table I.

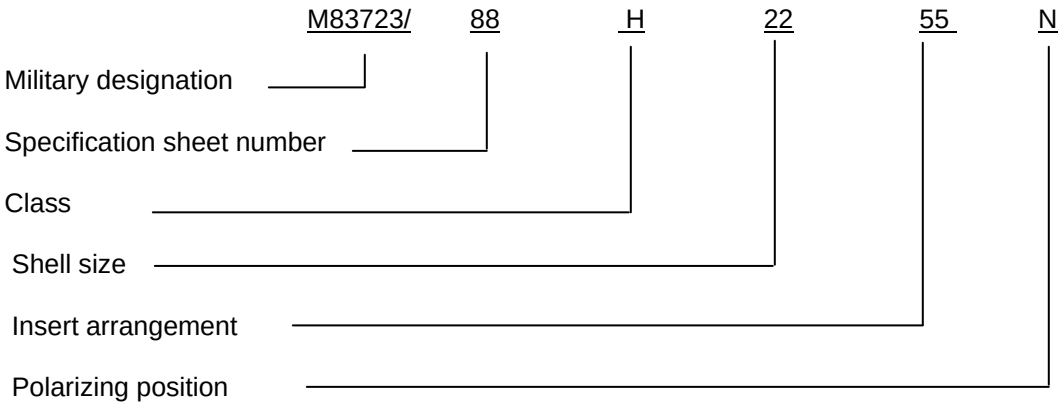
TABLE I. Insert arrangements requiring reduced diameters for raised seal barriers on contact cavities as indicated.

Shell size	Insert arrangement	Contact cavities
8	03, 98	1, 2, 3
10	06	1, 2, 3, 4, 5, 6

Connector accessories: See SAE-AS85049.

O-ring: See MIL-DTL-83723.

Part or Identifying Number (PIN) example:



Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-83723, this document references the following:

MIL-DTL-26500
MIL-DTL-83723/86
MIL-DTL-83723/91
MIL-DTL-83723/95
MIL-DTL-83723/97
MIL-STD-1554
MS3186
ASME-B46.1
ASTM B488
SAE-AS85049

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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – AS
Air Force – 85
DLA – CC

Preparing activity:
DLA – CC

(Project 5935-2010-052)

Review activities:
Army – AR, AV, MI
Navy – EC
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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